

The Hong Kong Polytechnic University

Subject Description Form

Please read the notes at the end of the table carefully before completing the form.

Subject Code	SFT546
Subject Title	Flexible Electronics for Wearable Technology
Credit Value	3
Level	5
Pre-requisite/ requisite/ Exclusion Co-	N/A
Objectives	The purpose of this subject is to help students understand the theme of flexible electronics, particularly for wearable applications, involving the selection of electronic components, processing technologies, major types of wearable electronic products, application scenarios, and a brief introduction of recent innovations. Through the study of this course, students will not only have a comprehensive understanding of the fundamental knowledge and principles of flexible electronics, but also be able to identify properties and applications of wearable electronic products as well as the latest development trends. Also, students will make full use of the knowledge and skills learned from this subject to analyze the possible problems in the current cases and think about the potential development priorities of flexible electronics in the industrial and academic fields in the future.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: a) Understand the characteristics of various functional materials, processing techniques and application scenarios of flexible electronics particularly in wearables; b) Access to the latest developments in flexible electronics to update their knowledge base; c) Understand the future development focus and potential application of flexible electronics ranging from artificial intelligence to smart living in the industrial and academic fields; d) Improve the comprehensive quality, including communication skills, reading ability, self-learning ability, teamwork ability, information integration capability, problem analyzing and solving ability. e) Develop mind, creativity and critical thinking.

Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	1. Electronic materials Electronic materials, mainly including inorganic materials (metals, inorganic semiconductors and carbon materials), organic materials (organic semiconductors and conducting polymer) and hybrids, are introduced, extending to electronic inks. 2. Processing technologies Types, features and equipment of processing technologies such as drop-casting, spraying, spinning and printing are involved. This section will focus on printing techniques. 3. Wearable electronic products Structures and functions of major types of products such as sensors, actuators, displays, antennas, energy harvesters and their hybrids are presented. Discussions will also be presented regarding limitations of products concerning manufacturability and performance as well as scientific understanding. Emphasis will be placed on fibre-based wearable electronic products and textile-based wearable electronic products. 4. Application scenarios Flexible electronics for wearable applications, including electronic skin, artificial intelligence, health-monitoring devices, etc.
Teaching/Learning Methodology <i>(Note 3)</i>	The lectures will deliver the fundamental knowledge and principles on materials, technologies, applications and progresses. Laboratory/studio is conducted to share the case studies, providing hands-on experience and deepening theoretical understanding. Students will be encouraged to communicate and discuss with each other or with the lecturer, and it is advocated to do a lot of extracurricular reading around their own topics and share their thoughts. Group projects and individual integrated design projects will be the major parts of continuous assessment components.

Assessment Methods in Alignment with Intended Learning Outcomes <i>(Note 4)</i>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	Continuous assessment	100%						
	Group Project–Report & Presentation (including comment and Q&A section)	40%	✓	✓	✓	✓	✓	
	Individual Integrated Design Project – Report & Poster	60%	✓	✓	✓	✓	✓	
	Total	100 %						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Group Project-Report & Presentation (40%): The group project will assess students’ classroom and extracurricular learning outcomes, mainly including the understanding of fundamental knowledge and the thinking and discussion of the whole subject. Individual Integrated Design Project-Report & Poster (60%) The individual integrated design project will appraise students’ comprehensive quality including the mastery of fundamental knowledge and skill, the accumulation of extracurricular reading, and the ability to analyze and solve problems. Students’ professionalism, including communication skills, presentation abilities, and development of critical and creative thinking, will also be assessed. The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.							
Student Study Effort Expected	Class contact:							
	▪ Lecture/ Seminar						26 Hrs.	
	▪ Studio/Laboratory						12 Hrs.	

	Other student study effort:	
	▪ Preparation for presentations/projects	30 Hrs.
	▪ Reading and writing	40 Hrs.
	Total student study effort	108 Hrs.
Reading List and References	Recommended Textbook Required readings 1. Zheng Cui, Printed electronics: materials, technologies and applications, Wiley, 2016. 2. Xiaoming Tao, Handbook of smart textiles, Springer, 2015. (e-version available) 3. D Tilak, Electronic textiles: smart fabrics and wearable technology, Woodhead, 2015. 4. William S. Wong and Alberto Salleo, Flexible electronics: materials and applications, Springer, 2009. Supplementary 1. Huisheng Peng, Xuemei Sun, Wei Weng, Xin Fang, Polymer materials for energy and electronic applications, Elsevier, Inc. 2017. (e-version available). 2. Abdelsalam (Sumi) Helal, Mounir Mokhtari, Bessam Abdulrazak, The Engineering Handbook of Smart Technology for Aging, Disability, and Independence, John Wiley & Sons, Inc. 2008. (e-version available) 3. Xiaoming Tao, Wearable electronics and photonics, Elsevier, 2005. Periodicals and Websites Advanced Materials Chemical Society Reviews Chemical Reviews Smart Materials and Structures Textile Research Journal Textile Outlook International	

Note 1: Intended Learning Outcomes

Intended learning outcomes should state what students should be able to do or attain upon subject completion. Subject outcomes are expected to contribute to the attainment of the overall programme outcomes.

Note 2: Subject Synopsis/Indicative Syllabus

The syllabus should adequately address the intended learning outcomes. At the same time, overcrowding of the syllabus should be avoided.

Note 3: Teaching/Learning Methodology

This section should include a brief description of the teaching and learning methods to be employed to facilitate learning, and a justification of how the methods are aligned with the intended learning outcomes of the subject.

Note 4: Assessment Method

This section should include the assessment method(s) to be used and its relative weighting, and indicate which of the subject intended learning outcomes that each method is intended to assess. It should also provide a brief explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes.